

Stripe

As a Business Analyst on the Stripe Capital team, you are tasked with evaluating the effectiveness of lending products for small businesses....

Question 1: What is the average monthly revenue for small businesses that received a loan versus those that did not receive a loan during January 2024? Use the "business_size" field to filter for small businesses.

```
WITH total AS(
  SELECT DISTINCT fct_loans.business_id, monthly_revenue,
    CASE WHEN loan_id IS NULL THEN 'Did Not Receive Loan'
      ELSE 'Received Loan' END AS Loan_Status
  FROM dim_businesses
  LEFT JOIN fct_loans
  ON fct_loans.business_id = dim_businesses.business_id
    AND strftime('%Y-%m', loan_issued_date) = '2024-01'
  WHERE business_size = 'small')

SELECT Loan_Status, AVG(monthly_revenue)
FROM total
GROUP BY Loan_Status
```



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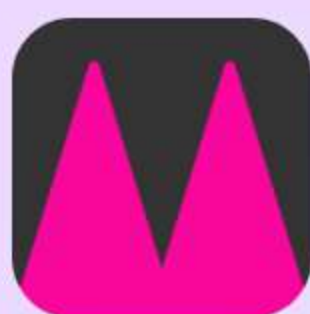
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Question 2: For loans issued to small businesses in January 2024, what percentage were successfully repaid? Categorize these loans based on the borrowing business's revenue variability (low, medium, or high) using these values
- Low: <0.1
- Medium: $0.1 - 0.3$ inclusive
- High: >0.3

```
WITH classify AS(
  SELECT business_id, revenue_variability,
         CASE WHEN revenue_variability < 0.1 THEN 'Low'
              WHEN revenue_variability > 0.3 THEN 'High'
              ELSE 'Medium' END AS revenue_group
  FROM dim_businesses
  WHERE business_size = 'small'
),

combine AS(
  SELECT DISTINCT(fct_loans.business_id), revenue_group, loan_repaid
  FROM fct_loans
  INNER JOIN classify
  USING (business_id)
  WHERE strftime('%Y-%m', loan_issued_date) = '2024-01')

SELECT
  revenue_group,
  COUNT(CASE WHEN loan_repaid = 1 THEN 1 END) AS repaid_count,
  COUNT(*) AS total_count,
  ROUND(COUNT(CASE WHEN loan_repaid = 1 THEN 1 END) * 100.0
        / COUNT(*), 2) AS repayment_percentage
FROM combine
GROUP BY revenue_group;
```



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Question 3: For small businesses during January 2024, what is the loan repayment success rate for each revenue variability category? Order the results from the highest to the lowest success rate to assess the correlation between revenue variability and repayment reliability.

```
with calculate AS(
  SELECT business_id, revenue_variability,
    CASE WHEN revenue_variability < 0.1 THEN 'Low'
         WHEN revenue_variability > 0.3 THEN 'High'
         ELSE 'Medium' END AS revenue_category
  FROM dim_businesses
  WHERE business_size = 'small')

SELECT revenue_category,
  100.0 * SUM(CASE WHEN loan_repaid = true THEN 1 ELSE 0 END)
/ COUNT(*) AS loan_repayment_success_rate
FROM calculate
INNER JOIN fct_loans
ON calculate.business_id = fct_loans.business_id
WHERE TO_CHAR(loan_issued_date, 'YYYY-MM') = '2024-01'
GROUP BY revenue_category
ORDER BY 2 DESC
```

